```
import java.util.*;
import javax.crypto.*;
import javax.xml.bind.DatatypeConverter;

public class lab444 {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

         //Generating a key
        KeyGenerator kg = KeyGenerator.getInstance("DES");
        SecretKey key = kg.generateKey();

        Cipher c = Cipher.getInstance("DES");
```

```
System.out.print("Enter message: ");
String msg = sc.nextLine();

c.init(Cipher.ENCRYPT_MODE, key);
byte[] encrypted = c.doFinal(msg.getBytes("UTF-8"));

String code = DatatypeConverter.printBase64Binary(encrypted);

c.init(Cipher.DECRYPT_MODE, key);
byte[] decrypted = c.
        doFinal(DatatypeConverter.parseBase64Binary(code));

System.out.println("Encrypted message: " + code);
System.out.println("Decrypted message: " + new String(decrypted, "UTF-8"));
```

```
        sc.close();
    }
}
```